

family buildings—to fulfil a definite set of power requirements. In commercial and industrial sectors, these can be used in places with chronic power shortage, or where the grid has not reached—like rural offices, schools or small hotels for basic needs—as only a few loads like fans and lights have to be powered. Due to this, standalone hybrid systems are generally preferred over on-grid systems.

Other application areas include military and railways. Indian Railways, for example, is already issuing tenders to implement wind-solar hybrid projects.

In spite of the potential of hybrid systems, a high initial investment, considerable duration for payback,

and uncertainty about their applicability makes people skeptical regarding their viability, thereby decreasing the rate of adoption and implementation in various countries. To bring a positive change, several companies in India are working in the area of wind-solar hybrid, including Supernova Technologies Private Limited, Zenith Solar Systems, Alpha Windmills, Unitron Energy System Pvt Ltd, and ReGen Powertech Pvt Ltd. The Ministry of New and Renewable Energy (MNRE) has introduced the National Wind-Solar Hybrid Policy to promote large-scale wind-solar hybrid power projects through measures like encouragement to technological advancements and better utilisation

of transmission infrastructure.

Outside India too, various initiatives are being taken to boost the sector. Gamesa, Blue Pacific Solar Products, Grupo Dragon, and Alternate Energy Company are some of the popular names in this field. Last year, the Kenyan Investment Authority and Meru County Government entered into a Memorandum of Understanding (MoU) with Windlab and Eurus Energy for the development of the Meru County Energy Park, African continent's first large-scale wind-solar-storage hybrid project. The aim is to produce up to 80MW of clean energy to power homes, and construction is expected to begin next year.

—Ayushee Sharma

3 INNOVATIONS THRIVING IN THE BATTLE AGAINST CORONAVIRUS

Amidst the outbreak of Covid-19 pandemic, existing technologies and innovations like robots and drones are becoming common. These are being used for sanitisation, awareness, adherence to social distancing rules, and to deliver medical supplies

Unforeseen situations call for unforeseen measures. Such is the case of the Covid-19 pandemic, where the urgency of response has resulted in a host of innovations and proactive measures through collaborations among universities, industries, startups, and the government.

Existing technologies like robots and drones are becoming common. Robots are being used round the clock in hospitals to provide medication and meals to patients. In office entrances and public places, these are being used for sanitisation and awareness. Drones are being used to monitor adherence to social distancing rules, do temperature screening using thermal imaging, and deliver medical supplies

Big data and availability of digital resources on government platforms are not only helping citizens but also those developing solutions by serving as an authentic source of information. In South Korea, artificial intelligence (AI) has been used to develop testing kits and other tools. For example, VUNO's Chest X-Ray AI image support decision tool uses an algorithm to clas-

sify intensive care patients by using X-ray images and can examine the lung within just three seconds.

Singapore, which in spite of being a small country is densely populated and relies on an open economy, has been able to manage well till now and kept its Covid-19 cases under control. Technological innovations and modification of pre-existing solutions to cater to the needs in this scenario have played a vital role in ensuring this.

For example, an automated thermal screening device iThermo scans people's temperature as they walk by at IHiS' office and St. Andrew's Community Hospital in Singapore and alerts staff when a person with a high body temperature is detected. People don't have to stop in queues, and frontline workers have lower chances of contact with the undetected person. The benefits of using such technologies include a lower risk of infection and reduced need for manpower in places like hospitals, according to *luxresearchinc.com*.

In India, innovations that can deal with multiple patients at a time are helping in dealing with the situation,

given the huge population of the country. One such example is Telemetric Patient Monitoring System by Cardiac Design Labs that allows healthcare workers and volunteers to monitor the vitals (like respiration, body temperature, and blood pressure) of multiple patients at a time from a central location by attaching wearable devices to their chests. Besides hospitals, it can also be used in homes, hotels, schools, and the like.

A large population of India lives in rural areas. Numerous startups are developing economical and portable ventilators for critical care owing to their shortage that can be deployed in both urban and rural areas of the country after medical regulatory approval.

AI and machine learning (ML) are playing a key role in helping state governments. For instance, in April, Bihar permitted the development of an AI-powered tool through a partnership between Norway India Partnership Initiative (NIPI) and Wadhvani Institute of AI that can determine potential patients through cough-sound analysis. In Andhra Pradesh, an ML-based